

ScheduleDesk: Complete App Guide

Master Data → Curriculum → Timetable Building

Multi-Year Timetable Creation Guide

4th, 3rd, 2nd, 1st Year CSM Sections

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1. App Navigation Overview

The left sidebar shows these main sections:

Section	Purpose	What You Do Here
Dashboard	Home	See all sections + their build status (what to do next)
Master Data	Setup foundation	Add Departments, Branches, Sections, Rooms, Subjects, Faculty
Term Setup	Configure timing	Create academic terms, set daily schedule (9am-5pm, periods, breaks)
Curriculum	Define workload	Add subjects to each section + required theory/lab hours/week
Faculty Assignment	Assign teachers	Pick which faculty teaches which subject in each section
Section Builder	Place classes	Manually drag-and-drop subjects into the timetable grid (clash-blocked)
Timetables	View results	See the final timetable for each section + print
Faculty Workload	Check balance	See each teacher's schedule + total hours/week
Import	Bulk load	Upload an Excel sheet to import an existing timetable
Print All	Export	Print/PDF all section timetables in one go
Reset Year	Archive old	Roll over to a new academic year (old data stays intact)

2. Dashboard & Build Status

When you first open the app, you see the Dashboard. This is your control center.

What the Dashboard shows:

- A table listing every section (4th Year CSM, 3rd Year CSM, etc.)
- For each section, a colored status badge showing the current stage:

Status Badge	Meaning	What to do next
needs-room	No home room set	Go Master Data → Sections → pick a room (the classroom where theory classes happen)
needs-curriculum	No subjects assigned	Go Curriculum → click the section → add subjects + required weekly hours
needs-faculty	Subjects exist but no faculty assigned	Go Curriculum → click section → click each subject → pick faculty from dropdown
ready-to-build	All setup complete	Go Section Builder → start placing subjects into the timetable grid
in-progress	Some subjects placed	Go Section Builder → continue placing remaining subjects
done	All subjects placed, zero conflicts	Go Timetables → view the complete timetable → Print

Key insight: The Dashboard tells you exactly what to do next for each section. Just follow the status badge → click the linked section → complete that step. Repeat for all sections.

3. Master Data Management

Master Data is the foundation. Enter it ONCE, and it's reusable across all years.

What to enter in Master Data:

- Departments (CSE, ECE, Mech, etc.)
- Branches (CSM, CAI, CSD, AIML under each dept)
- Sections (Year 1 CSM, Year 2 CSM, Year 3 CSM, Year 4 CSM)
- Rooms (classrooms, labs, seminar halls, library)
- Subjects (Machine Learning, Deep Learning, etc.)
- Faculty (all teachers + their department)

Pro tip: Set up Departments and Branches ONCE. Then for each year level, just create a new Section row (e.g., "1st Year CSM", "2nd Year CSM", etc.). You reuse the same Faculty, Subjects, and Rooms for all years.

4. Academic Term Setup

A term defines the academic year and the daily class timing.

Create ONE term per year:

- Go Term Setup → Create New Term
- Enter: Year (2024), Semester (1 or 2), Start Date, End Date
- Set it as Active (only one term can be active at a time)
- Configure Time Config (this is the daily schedule)

Time Config (Daily Schedule):

This tells the app when classes start, how long each period is, and where breaks/lunch fall. It's applied to all sections in that term. Example:

- Start: 09:00 (classes start at 9 AM)
- Periods: 7 (the day has 7 teaching periods)
- Period Length: 50 minutes (each class is 50 min)
- Break After Period 2: 15 minutes (short break after 2nd period)
- Lunch After Period 4: 50 minutes (lunch break after 4th period)
- Result: Full day = 9:00 AM to 5:55 PM

Pro tip: If you want a different schedule later (e.g., 9:00–4:00 with different break times), create a new Term with a different TimeConfig. Don't change the active term's config mid-year.

5. Curriculum Management

Curriculum = the list of subjects taught in a section + how many hours/week each requires. This is where you define the workload.

For each section, add subjects:

1. Go Curriculum → pick the section (e.g., 4th Year CSM)
2. Click 'Add Subject'
3. Choose a subject from the dropdown (e.g., Machine Learning)
4. Enter Theory Hours/Week (e.g., 4) and Lab Hours/Week (e.g., 0)

5. Save

6. Repeat for all subjects in that section

Important: The total weekly hours you enter here **MUST** be achievable in the timetable. The app checks this at the end — if you promise 4 hours of Machine Learning/week but only place 3 hours worth of classes, it will stop you from saving until you fix it.

Then assign faculty:

1. In the same Curriculum screen, click each subject

2. A dropdown shows Faculty → pick who teaches this subject

3. The faculty must be marked as eligible for that subject (in Master Data → Faculty)

4. Save

Pro tip: Dr. Sai Kumar teaches both Machine Learning and Deep Learning? Add him to both subjects' eligibility (Master Data → Faculty → click Sai → Add Subjects: ML, DL). Then in Curriculum, you can assign him to both.

6. Timetable Building (The Fun Part)

Once Curriculum is complete (status = 'ready-to-build'), you go to Section Builder to manually place subjects.

How to build a timetable:

1. Go Section Builder → pick a section (e.g., 4th Year CSM)
2. You see a grid: columns = periods (P1-P7), rows = days (Mon-Fri)
3. Pick a subject from the dropdown at the top (e.g., Machine Learning)
4. The grid shows only LEGAL slots (green/white). Illegal slots are greyed out with a reason.
5. Click a legal slot to place the subject there
6. A subject taking 4 hours/week = place 4 × 50-minute slots
7. Labs = 3 continuous periods in one click (they auto-snap to valid windows)
8. Repeat until all subjects are placed

What the app prevents (clash detection):

- Faculty clash: A teacher can't teach two classes at the same time
- Section clash: A section can't have two classes at the same time
- Room clash: The home room can't have two sections at once
- Lab window rules: Labs can span lunch but NOT the break (break stops continuity)
- Weekly hour mismatch: You can't save a timetable that under- or over-delivers hours

Pro tip: If a slot is greyed out, hover over it to see WHY it's illegal. This helps you understand the rules and find the next legal slot.

7. How to Delete/Clear a Timetable

Want to start over for a section? Or clear all entries to rebuild? Here's how:

Option A: Clear a Single Entry (Remove One Class)

1. Go Section Builder → pick the section 2. Find the class you want to remove (hover over the cell to see the subject name) 3. Click the cell → a delete button appears 4. Click delete 5. The slot is now free and shows in the grid again

Option B: Clear the Entire Section's Timetable

1. Go Section Builder → pick the section 2. Look for a "Clear All" or "Reset Section" button (usually at the top right) 3. Confirm (it will ask 'Are you sure?') 4. All placed entries for this section are deleted 5. The curriculum stays intact — you just removed the placement schedule

Option C: Delete a Section Entirely (Nuclear Option)

1. Go Master Data → Sections 2. Find the section you want to delete (e.g., 4th Year CSM - A) 3. Click the delete icon 4. Confirm 5. The section + its curriculum + its timetable are all gone (can't undo!)

Warning: Deleting a section is permanent and removes EVERYTHING (curriculum, assignments, timetable). Only do this if the section truly doesn't exist. To temporarily hide a section, leave its timetable empty instead.

8. Multi-Year Workflow (4th, 3rd, 2nd, 1st Year)

Now you'll create timetables for ALL year levels. The process is the same for each, but here's an optimized workflow to save time:

ONCE (at the start of the year):

1. Master Data: Add all Departments, Branches (CSE, ECE, etc. with CSM, CAI, etc.)
2. Master Data: Add all Rooms (list all classrooms, labs, etc. for the college)
3. Master Data: Add all Subjects (all possible courses taught anywhere in the college)
4. Master Data: Add all Faculty (all teachers + their department)
5. Master Data: Mark Faculty Eligibility (which teacher can teach which subjects)
6. Term Setup: Create one Academic Term with the daily TimeConfig

FOR EACH YEAR LEVEL (repeat this for 4th, 3rd, 2nd, 1st):

1. Master Data → Sections: Create a new section: "4th Year CSM" (or "3rd Year CSM", etc.)
2. Pick the Branch (CSM) and Home Room (the classroom where theory classes happen)
3. Curriculum → Pick "4th Year CSM": Add all subjects required for 4th year + weekly hours
4. Curriculum → Assign Faculty: Click each subject → pick the teacher
5. Status changes to 'ready-to-build' automatically
6. Section Builder → Pick "4th Year CSM": Manually place all subjects into the grid
7. Once all subjects placed with zero conflicts, status = 'done'
8. Repeat for "3rd Year CSM", "2nd Year CSM", "1st Year CSM"

AT THE END:

1. Go Timetables → View each section's completed grid
2. Go Faculty Workload → Verify no teacher is overloaded
3. Go Print All → Save one PDF with every section's timetable

Step-by-Step Example: Creating All Four Years

YEAR-BY-YEAR CHECKLIST

■ YEAR 4 (4th Year CSM)

Step	Action	Status Change
1	Master Data → Sections: Create "4th Year CSM", Branch=CSM, HomeRoom=Lab-A1	
2	Curriculum → Add subjects (ML 4hrs, DL 3hrs, DLL 3hrs lab, etc.)	needs-curriculum → needs-faculty
3	Curriculum → Assign Faculty (Sai to ML, Anu to DL Lab, etc.)	needs-faculty → ready-to-build
4	Section Builder → Place all subjects into grid	ready-to-build → in-progress
5	Verify timetable → Go Timetables → 4th Year CSM (view grid)	in-progress → done

■ YEAR 3 (3rd Year CSM)

Step	Action	Status Change
1	Master Data → Sections: Create "3rd Year CSM", Branch=CSM, HomeRoom=Lab-A1	
2	Curriculum → Add subjects (AI 4hrs, ML 3hrs, ADS 2hrs, Big Data 3hrs lab, etc.)	needs-curriculum → needs-faculty
3	Curriculum → Assign Faculty	needs-faculty → ready-to-build
4	Section Builder → Place subjects	ready-to-build → done
5	Verify timetable	done

■ YEAR 2 (2nd Year CSM)

Step	Action	Status Change
1	Master Data → Sections: Create "2nd Year CSM", Branch=CSM, HomeRoom=Lab-A1	
2	Curriculum → Add subjects (DSA 4hrs, OOP 3hrs, DB 3hrs lab, Web Dev 2hrs, etc.)	needs-curriculum → needs-faculty
3	Curriculum → Assign Faculty	needs-faculty → ready-to-build
4	Section Builder → Place subjects	ready-to-build → done
5	Verify timetable	done

■ YEAR 1 (1st Year CSM)

Step	Action	Status Change
1	Master Data → Sections: Create "1st Year CSM", Branch=CSM, HomeRoom=Lab-A1	
2	Curriculum → Add subjects (C/C++ 3hrs, Unix 2hrs, Maths 4hrs, DSA Intro 2hrs lab, etc.)	needs-curriculum → needs-faculty
3	Curriculum → Assign Faculty	needs-faculty → ready-to-build
4	Section Builder → Place subjects	ready-to-build → done
5	Verify timetable	done

9. Common Errors & Solutions

Error: "Cannot save timetable — weekly hours mismatch"

This means you promised more (or fewer) hours in Curriculum than what you actually placed in the timetable. Solution: Check Curriculum for this section. E.g., if you promised 4 hours of ML/week but only placed 3 × 50-minute slots, either add one more ML slot or reduce the curriculum to 3 hours.

Error: "Faculty clash — Dr. Sai is already teaching at this time"

The slot is greyed out because this faculty member is already assigned to another class at that time. Solution: Pick a different time slot, or check if Dr. Sai is teaching another section at the same time (go Faculty Workload to see all their classes).

Error: "Invalid lab window — labs must be 3 consecutive periods"

Labs must occupy exactly 3 continuous periods. The slot you clicked won't fit 3 periods before the day ends or the break blocks continuity. Solution: Click a different period. The app highlights valid lab-start periods in green. Labs can span lunch (lunch doesn't break them) but NOT the break.

Error: "Section needs a home room before you can place classes"

The status badge said 'needs-room', meaning you didn't pick a classroom for this section. Solution: Go Master Data → Sections → click the section → pick a Home Room from the dropdown (e.g., Lab-A1). Then go back to Section Builder.

Error: "Faculty not assigned to this subject"

You're trying to place a subject, but no faculty member is assigned to teach it in this section. Solution: Go Curriculum → click the section → click the subject → assign a faculty from the dropdown. The faculty must be marked as eligible for that subject first.

Error: "I accidentally deleted all classes — can I undo?"

There's no undo button. Once deleted, the entries are gone. Solution: If this just happened, go Section Builder → place them again (faster the second time, you know where they go). If this happened hours ago and you don't remember, consider using the Import feature to re-upload the old Excel sheet.

Pro Tips for Smooth Workflow

- Do Master Data ONCE at the start of the year. Then reuse everything for all sections.
- Follow the status badges on the Dashboard. They tell you what to do next.
- When placing classes, start with the hardest subjects (those taught by few faculty, with fixed times). Then fill around them.
- Use Faculty Workload frequently to check if anyone is over-booked before saving.
- If two sections need the same faculty at the same time, the app will block it. Plan accordingly.
- Print All produces one PDF with every section. Use this to distribute to faculty.
- If a term changes in the middle (e.g., new exam dates), use Reset Year to roll into a new term instead of editing the active one.